INE Skill Check Lab Report
Assessment Methodologies: Footprinting and Scanning CTF 1

Target: target.ine.local
Assessment Type: Footprinting & Scanning / Service Enumeration
Tools Used: Nmap, enum4linux, smbclient, Metasploit, FTP, SSH

Flags Captured

Flag	Value
Flag 1	FLAG1{ac0b2ab046944fa9a126aa0c0661d1ee}
Flag 2	FLAG2{22ae10eb25f54b008273abe9938f83be}
Flag 3	FLAG3{c36d93d53b4a48be8252ae55ec73f022}
Flag 4	FLAG4{b50f96fac09c489080d3b8bd9284527c}

Flag 1 – Anonymous SMB Share

The first flag was obtained from an anonymously accessible Samba share named pubfiles. Using smbclient, the share was accessed without authentication and the file flag1.txt was downloaded.

```
┌──(root㉿INE)-[~/Desktop/wordlists]
└─# smbclient //target.ine.local/pubfiles -N
Try "help" to get a list of possible commands.
smb: \> ls
  .                                   D        0  Fri Mar 13 02:39:25 2026
  ..                                  D        0  Tue Nov 19 10:44:41 2024
  flag1.txt                           N       40  Fri Mar 13 02:39:25 2026

                2112647088 blocks of size 1024. 177450056 blocks available
smb: \> get flag1.txt
getting file \flag1.txt of size 40 as flag1.txt (39.1 KiloBytes/sec) (average 39.1 KiloBytes/sec)
smb: \> ^Z
[1]+  Stopped                 smbclient //target.ine.local/pubfiles -N

┌──(root㉿INE)-[~/Desktop/wordlists]
└─# cat flag1.txt
FLAG1{ac0b2ab046944fa9a126aa0c0661d1ee}
```

Flag 2 – Weak SMB Credentials

Enumeration revealed several SMB users. A brute force attempt using Metasploit identified weak credentials for the user josh with password purple. After authentication, the private share 'josh' was accessed and flag2.txt was retrieved.

```
┌──(root㉿INE)-[~/Desktop/wordlists]
└─# smbclient //target.ine.local/josh -U josh
Password for [WORKGROUP\josh]:
Try "help" to get a list of possible commands.
smb: \> ls
  .                                   D        0  Fri Mar 13 02:39:25 2026
  ..                                  D        0  Tue Nov 19 10:44:41 2024
  flag2.txt                           N      119  Fri Mar 13 02:39:25 2026

                2112647088 blocks of size 1024. 177415972 blocks available
smb: \> get flag2.txt
getting file \flag2.txt of size 119 as flag2.txt (116.2 KiloBytes/sec) (average 116.2 KiloBytes/sec)
smb: \> ^Z
[3]+  Stopped                 smbclient //target.ine.local/josh -U josh

┌──(root㉿INE)-[~/Desktop/wordlists]
└─# cat flag2.txt
FLAG2{22ae10eb25f54b008273abe9938f83be}

Psst! I heard there is an FTP service running. Find it and check the banner.

┌──(root㉿INE)-[~/Desktop/wordlists]
└─#
```

Flag 3 – FTP Service Enumeration

A hint in Flag 2 suggested an FTP service was running. Service scanning identified FTP running on port 5554. After connecting to the FTP service and logging in, the file flag3.txt was found and downloaded.

```
┌──(root㉿INE)-[~/Desktop/wordlists]
└─# ftp target.ine.local 5554
Connected to target.ine.local.
220 Welcome to blah FTP service. Reminder to users, specifically ashley, alice and amanda to change their weak passwords
Name (target.ine.local:root): alice
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||45322|)
150 Here comes the directory listing.
-rw-rw-r--    1 0        0              40 Mar 12 21:09 flag3.txt
226 Directory send OK.
ftp> get flag3.txt
local: flag3.txt remote: flag3.txt
229 Entering Extended Passive Mode (|||5732|)
150 Opening BINARY mode data connection for flag3.txt (40 bytes).
100% |*****************************************************************************************************************
226 Transfer complete.
40 bytes received in 00:00 (255.30 KiB/s)
ftp>
[11]+  Stopped                 ftp target.ine.local 5554

┌──(root㉿INE)-[~/Desktop/wordlists]
└─# cat flag3.txt
FLAG3{c36d93d53b4a48be8252ae55ec73f022}
```

Flag 4 – SSH Login Banner

The final flag was discovered by connecting to the SSH service. The login banner displayed a warning message to deter unauthorized users and contained the final flag.

```
┌──(root㉿INE)-[~/Desktop/wordlists]
└─# ssh target.ine.local
The authenticity of host 'target.ine.local (192.213.232.3)' can't be established.
ED25519 key fingerprint is SHA256:qWHJnmTFgrmLKFbmMNRLIr1Y8MVWpqGGxhJ5miFHgnQ.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'target.ine.local' (ED25519) to the list of known hosts.
*********************************************************************
*                                                                   *
*            WARNING: Unauthorized access to this system            *
*            is strictly prohibited and may be subject to           *
*            criminal prosecution.                                  *
*                                                                   *
*            This system is for authorized users only.              *
*            All activities on this system are monitored            *
*            and recorded.                                          *
*                                                                   *
*            By accessing this system, you consent to               *
*            such monitoring and recording.                         *
*                                                                   *
*            If you are not an authorized user,                     *
*            disconnect immediately.                                *
*                                                                   *
*********************************************************************
*                                                                   *
*    Is this what you're looking for?: FLAG4{b50f96fac09c489080d3b8bd9284527c}       *
*                                                                   *
*********************************************************************
root@target.ine.local's password:
```

Conclusion

This assessment demonstrates how simple enumeration techniques can reveal critical security weaknesses such as anonymous file shares, weak credentials, and exposed services. Proper configuration and stronger authentication mechanisms are essential to reduce the attack surface.